

homogeneous linear equations with constant coefficients

recall $\sqrt{b^2 - 4ac}$ this is called the discriminant

- 1) $b^2 - 4ac > 0 \Rightarrow$ two real solutions
- 2) $b^2 - 4ac = 0 \Rightarrow$ one real solution by repeating example: $(m-s)(m-s)$
- 3) $b^2 - 4ac < 0 \Rightarrow$ complex conjugate, you will have a complex number

to understand this chapter you will need to memorize these three cases and the equations

Case 1:

$$b^2 - 4ac > 0 \quad y = C_1 e^{m_1 x} + C_2 e^{m_2 x}$$

- when you have two different solutions you are dealing with Case # 1 you will plug the two solutions for m_1 and m_2

Case 2:

$$b^2 - 4ac = 0 \quad y = C_1 e^{m_1 x} + C_2 x e^{m_1 x}$$

- when you have two solutions that are the same you are dealing with a Case # 2 you will plug in the solutions for m_1 and m_1

Case 3:

$$b^2 - 4ac < 0 \quad y = e^{\alpha x} (C_1 \cos(\beta x) + C_2 \sin(\beta x))$$

- when you have a imaginary solution you are dealing with a Case # 3

$$\begin{array}{cc} \alpha + \beta i \\ \uparrow \quad \uparrow \\ \text{real} \quad \text{imaginary} \\ \text{part} \end{array}$$

$$\alpha = \text{alpha}$$

$$\beta = \text{beta}$$

ex:

Find the general solution of the 2nd-order D.E

$$a) \quad y'' - 3y' + 2y = 0$$

$$\Rightarrow m^2 - 3m + 2 = 0$$

• notice how this is a quadratic equation

$$(m-2)(m-1) = 0$$

$$m=2 \quad m=1$$

$$y = c_1 e^{m_1 x} + c_2 e^{m_2 x}$$

$$y = c_1 e^{2x} + c_2 e^x$$

• when you see y'' you can write it as m^2
when you see y' you can write it as m
but when you have a regular letter that is not a derivative like $3y$ you can only write it as 3

• this is a case #1 because it has two different solutions

$$b) \quad y'' - 10y' + 25y = 0$$

$$\Rightarrow m^2 - 10m + 25 = 0$$

$$(m-5)(m-5) = 0$$

$$m=5 \quad m=5$$

$$y = c_1 e^{m_1 x} + c_2 x e^{m_1 x}$$

$$y = c_1 e^{5x} + c_2 x e^{5x}$$

• this is a case #2 because the answers are both identical

$$c) \quad y'' - 4y' + 5y = 0$$

$$m^2 - 4m + 5 = 0$$

• you will need Quadratic Formula

$$m = \frac{-(-4) \pm \sqrt{16 - 4(5)}}{2(1)} = \frac{4 \pm 2i}{2}$$

$$\Rightarrow 2 \pm i$$

$$\begin{array}{cc} 2 \pm i \\ \uparrow \quad \uparrow \\ \alpha \quad \beta \end{array}$$

$$y''' + 3y'' + 3y' + y = 0$$

$$m^3 + 3m^2 + 3m + 1 = 0$$

Synthetic division

$$m^3 + 3m^2 + 3m + 1 = 0$$

↑ ↑
Q P

Find all the numbers that are divisible by Q and P

• divide all P's by Q

$$Q: \pm 1$$

$$P: \pm 1$$

$$\frac{P}{Q} = \pm 1$$

once you have done $\frac{P}{Q}$ you can plug into the equation until you get zero

$$f(1) \neq 0, \quad f(-1) = 0 \checkmark$$

this is r

multiply

-1	1	3	3	1	
	↓	-1	-2	-1	
	1	2	1	0	

add

• the answers is what you will place in front of the

• this is a case #3 because it has complex numbers in it

$$y = e^{\alpha x} (C_1 \cos(\beta x) + C_2 \sin(\beta x))$$

$$y = e^{2x} (C_1 \cos(x) + C_2 \sin(x))$$

★ β is one

• you will need to apply synthetic division to solve this

• you will multiply and what ever answer you get you will place that for the next step for addition if you end up with zero you did it correct

$$(x-r) (m^2 + 2m + 1)$$

$$(m+1) (m^2 + 2m + 1)$$

$$(m+1) (m+1) (m+1)$$

• this is a case #2 because all the answers are the same

$$y = C_1 e^{m_1 x} + C_2 x e^{m_1 x} + C_3 x^2 e^{m_1 x}$$

$$y = C_1 e^x + C_2 x e^x + C_3 x^2 e^x$$

$$y'' + 5y' + 6y = 0$$

$$m^2 + 5m + 6 = 0$$

$$(m+3) (m+2) = 0$$

$$m = -3, m = -2$$

• this is a case #1 because there are two different solutions

$$y = C_1 e^{m_1 x} + C_2 e^{m_2 x}$$

$$y = C_1 e^{-3x} + C_2 e^{-2x}$$

$$y''' - 4y'' - 5y' = 0$$

$$m^3 - 4m^2 - 5m = 0$$

$$\begin{array}{cc} \uparrow & \uparrow \\ Q & P \end{array}$$

$$P: \pm 1, \pm 5$$

$$Q: \pm 1$$

$$\frac{P}{Q} = \pm 1, \pm 5$$

• you need to use synthetic division

$$\begin{array}{r|rrrr} -1 & 1 & -4 & -5 & \\ & & -1 & 5 & \\ \hline & 1 & -5 & 0 & \end{array}$$

$$(x-r) (m^2 - 5m) = 0$$

$$(m+1) m(m-5) = 0$$

$$m = -1, m = 0, m = 5$$

• this is case #1 because they are all different solutions

$$y = C_1 e^{-x} + C_2 + C_3 e^{5x}$$

$$\frac{d^3 u}{dt^3} + \frac{d^2 u}{dt^2} - 2u = 0 \Rightarrow u^3 + u^2 - 2u = 0$$

$$u^3 + u^2 - 2u = 0$$

Q P

$$P = \pm 1, \pm 2$$

$$Q = \pm 1$$

$$\frac{P}{Q} = \pm 1, \pm 2$$

$$f(1) = 0 \checkmark$$

a cubic Polynomial looks like this:

$$au^3 + bu^2 + cu + d = 0$$

So we will need to add this:

$$u^3 + u^2 + 0u - 2 = 0$$

$$\begin{array}{r|rrrrr} 1 & 1 & 1 & 0 & -2 & \\ & \downarrow & & & & \\ & 1 & 2 & 2 & 0 & \end{array}$$

$$(u-1)(u^2+2u+2)$$

← apply quadratic formula

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\begin{aligned} a &= 1 \\ b &= 2 \\ c &= 2 \end{aligned}$$

$$\frac{-2 \pm \sqrt{2^2 - 4(1)(2)}}{2(1)}$$

→

$$\frac{-2 \pm 2i}{2}$$

→

$$-1 \pm i$$

$$u = 1$$

↓

case #1

$$u = -1 \pm i$$

↓

case #3

• you sometimes can have two cases in one

$$y = C_1 e^x + e^{-x} (C_2 \cos(x) + C_3 \sin(x))$$